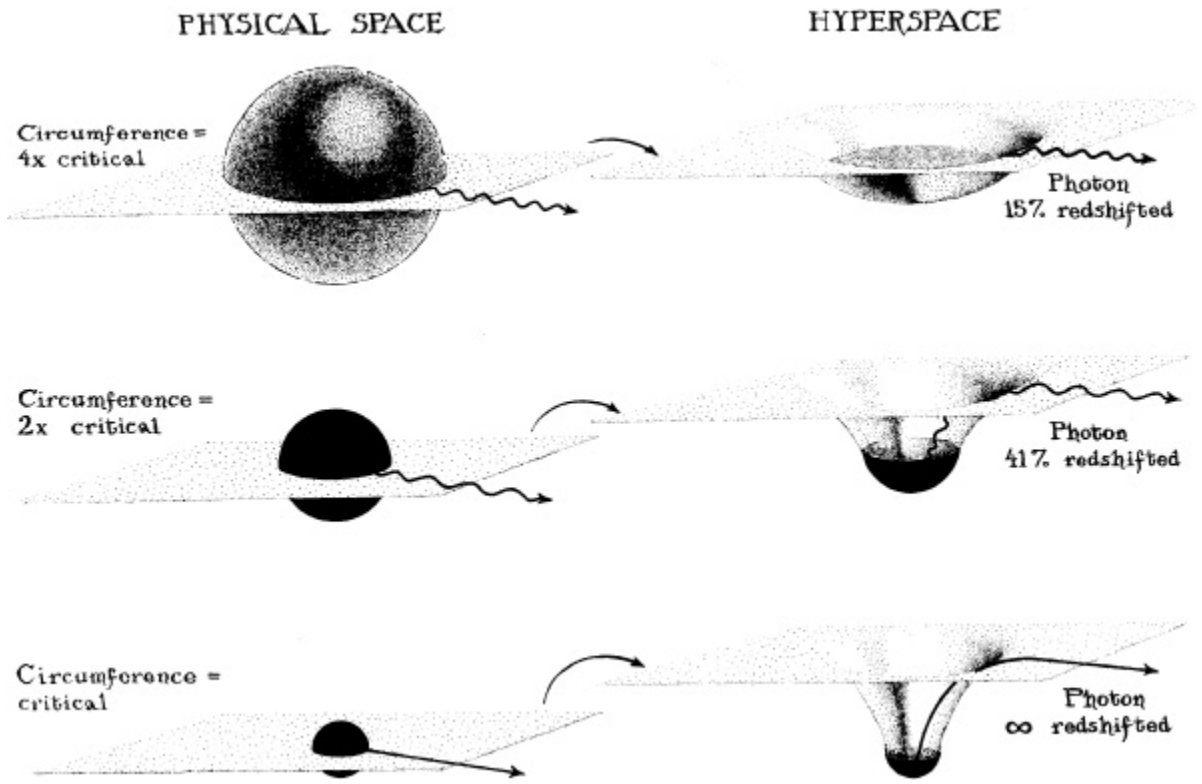


6.2 (Same as Figure 3.4.) General relativity's predictions for the curvature of space and the redshift of light from a sequence of three highly compact, static (non-imploding) stars that all have the same mass but have different circumferences.

The figure shows a sequence of three static stars that mimic the implosion that Oppenheimer and Snyder were preparing to analyze. Each star has the same mass, but they have different circumferences. The first is four times bigger around than the *critical circumference* (four times bigger than the circumference at which the star's gravity would become so strong that it forms a black hole). The second is twice the critical circumference, and the third is precisely at the critical circumference. The embedding diagrams show that the closer the star is to its critical circumference, the more extreme is the curvature of space around the star. However, the curvature does not become infinitely extreme. The bowl-like geometry is smooth everywhere with no sharp cusps or points or creases, even when the star is at its critical circumference; that is, the spacetime curvature is not infinite, and, correspondingly, since *tidal gravitational forces* (the kinds of forces that stretch one from head to foot and produce the tides on the Earth) are the physical manifestation of spacetime curvature, tidal gravity is not infinite at the critical circumference.